

A Thinking Web

Here, we look at the idea of software agents on the Web, or Web agents. We've mentioned agents in *The Lizard Of Oz, Digit*, January 2006. The Web has thus far been more or less a passive information repository—when you're looking for information, it's *you* that has to be intelligent; *you* need to supply the search engine with an intelligent query.

Enter the agent. Much like a bat that detects its prey by spreading echo waves and having the prey reflect them, you could let a Web agent “fan out” over the Web and have it return relevant information to you. Say you're looking for whether gorillas are apes; also, you want to list out what people think are the smartest animals, and whether gorillas figure in that list or not. This is a complex query, so you “weight” your search term: you give “gorillas” and “smart” a high weightage, “apes” and “animals” a lower weightage, and “list” the lowest weightage.

Your agent could make copies of itself and actually have a presence all over the Web, not knowing exactly what it's looking for, and come up with the best matches. For example, it might spot that a particular domain has interesting information on animal taxonomy and try and apply “ape” in that domain.

Just like regular software agents, Web agents could learn about their users. And agents could learn from each other, exchanging experiences!

This is not the space to analyse what human thinking consists in, but introspection will reveal that when we think about something—as when we try and solve a problem—we start off at a point, and fan our thoughts out to all the concepts and ideas we have. Our thoughts play the role of agents; the concepts and ideas play the role of the clusters we talked about earlier.

Thinking and learning could be combined. The knowledge an agent gains in its journeys across the Web could be passed on to the Web itself. And the learning Web would, in return, facilitate the agents' job.

For all this to happen, data on the Web needs to be structured better than it is now. It could happen in many ways, but one set of technologies waiting to take off comprise the Semantic Web (www.semanticweb.org), a vision of Sir Tim Berners-Lee, the inventor of the WWW. Berners-Lee, in *Weaving The Web*, 1999, said: “If HTML and the Web made all the online documents look like one huge book, (Semantic Web technologies) will make all the data in the world look like one huge database.” Documents on the Semantic Web are much more easily processed by machines than HTML documents are, and Web agents are more easily imaginable on something like the Semantic Web.

Eliciting Information

Now here's where it gets really interesting. The Overbrain could very well become a repository of all the world's information. Someone high up at Google did recently say that the company would index all the world's information by 2050—whether he said it in jest or not is not the point; the fact is, we're certainly moving in that direction. But our point is also that this repository would be a *smart* repository in a number of ways:



On the PCP Web, algorithms will reinforce popular links, while rarely used links will diminish and die. It's the first step on the road to the global brain

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- ☑ It would be “in touch” with human experts, who would be nodes on it.
- ☑ It would maintain the consistency of its knowledge, implying that, while being distributed, there would be a centralised check. Think of your brain: you cannot possibly harbour two entirely contradictory beliefs!
- ☑ To this end, it would elicit information from the experts and from its users.

To elucidate, interaction with the Overbrain need not be one-sided—after all, in the case of our brains, information flows both ways. The Overbrain could keep a check on the consistency of the knowledge it contained; for example, it should not be the case that one node says the plural of “virus” is “viruses,” with another saying it's “virii.” This is a silly example, but suppose the Overbrain detected such a mismatch. It could consult an expert in biology (or software) and ask for his opinion—using which it would update the data in the mistaken node. This is a massive step: if this were to actually come to pass, the Internet would be 100 per cent reliable, and this alone would make it a hundred times more useful.

The natural question now is, how would an expert be located? We can envisage a scenario in which experts register themselves with the system. Every individual has knowledge unique to him, and that others can benefit from. As more and more people plug in to the Overbrain, we'd have what we like to call a techno-ideosphere. Each one of us would have some form of identification. The domain and level of knowledge that one has to offer would be encoded in the identification, and there could be a sort of “global login” procedure. We can then envisage a system that offers knowledge in return for knowledge—so you log in as a gardening expert, and take help on subjects such as sports. We can also think of knowledge flowing between experts, resulting in debates and discussions that will enrich the usefulness of the Overbrain.

Memes On The Internet

Memes are ideas that can be passed on to others. In *The Meme Machine*, Susan Blackmore, psychologist and memeticist, has suggested that humans are basically carriers for memes—just like anthropologist Desmond Morris suggested that we are merely carriers for genes. An example of a meme is Darwin's idea that we descended from the apes. The Internet is, as we can all see, amazingly suited to the propagation of memes. Many memes start off at blogs. An example of a meme currently doing the rounds is the anti-evolutionary idea of Intelligent Design (for more information, visit www.arn.org). This is a good example because a few months ago, nobody had heard about it, and all of a sudden, the Web is abuzz with the phrase.

How do memes tie in with what we've been saying about the Overbrain? Well, the Overbrain can be a formal facilitator of meme exchange. Take the case of Idea-X, at <http://idea-x.net>. From phptr.com, “Idea-X is an online idea exchange. Participants can either propose ideas or ask for ideas to address a specific problem. A suite of tools allows people to see how other members rate each of the ideas and the people proposing them, and to keep track of the best ideas on the site.”